

Vertebral Column and Spinal Cord

Department Anatomy
Neuroanatomy (4)

Second Part

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Lectuer Note:

➤ Introduction and Review of The vertebral column.

- The vertebral column.
- Vertebrae
- Gneral featuers of vertebrae.
- Curvatuers of vertiberal colum.
- Cervical Vertebrae.
- First and Second Cervical Vertebrae.
- The Thoracic Vertebrae.
- The lumber vertebrae.
- The Joints between vertebrae.
- Ligaments between vertebrae.
- The disc.
- Blood supply of vertebra and spinal cord.
- Nerve Supply of Vertebral Joints.
- Dermatomes innnervated by posterior rami of spinal nerves
- Intervertebral foramen

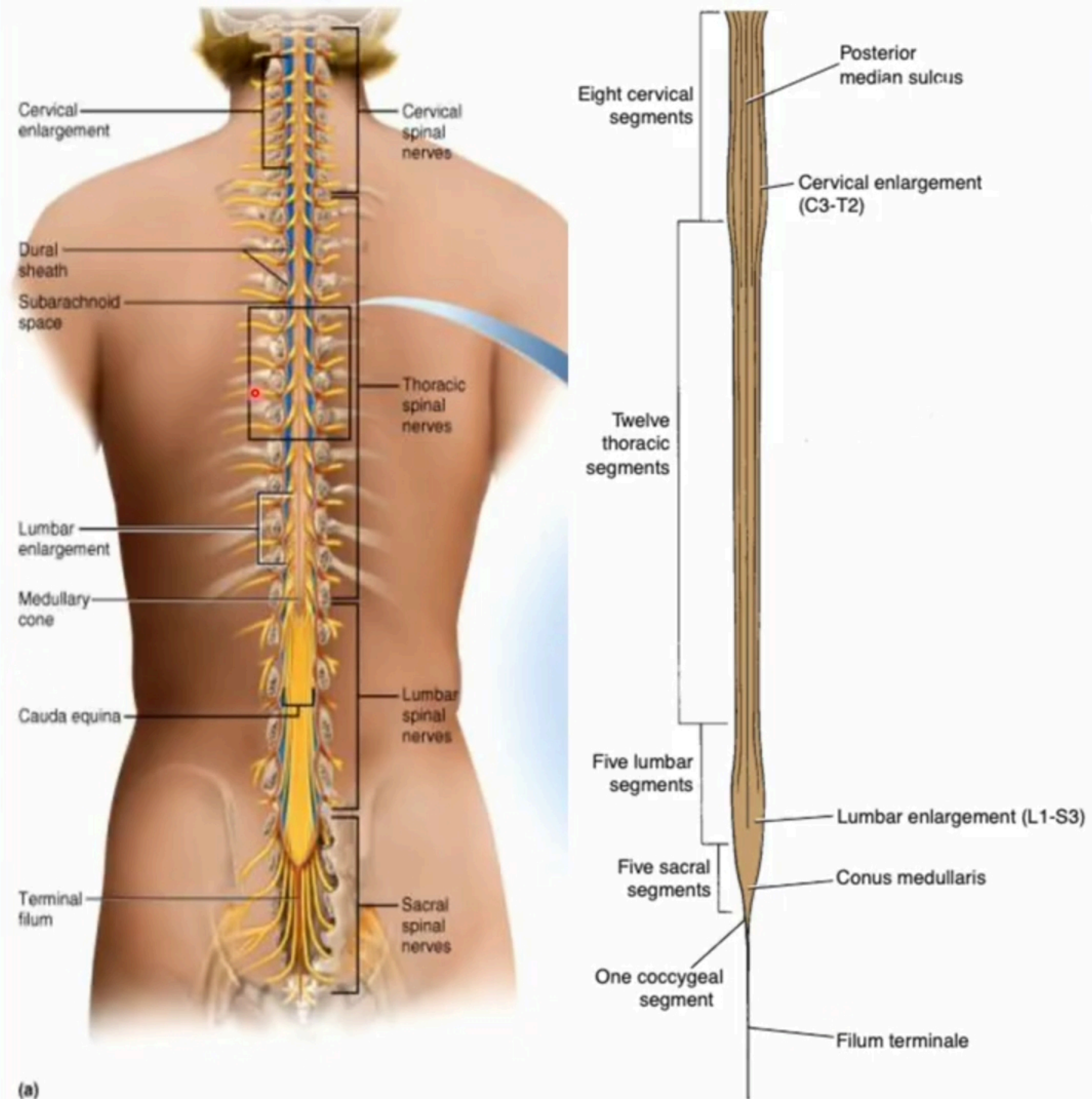
➤ The Spinal Cord

- Introduction of Spinal Cord.
- Lumbar Puncture (Spinal Tap).
- Curvatuers of vertiberal colum deferate end of spinal cord between adult and infant and young child.
- Function of spinal cord.
- Spinal cord has:
 - Internal Featuers.
 - External Features.
- Spinal Meninges amd spaces.
- Spinal Segments.
- The gray matter of the spinal cord.
- The White matter of Spinal Cord
- Transverse sections of the spinal cord at different levels showing the arrangement of the gray matter and white matter.

- Important notes about the white mater of Spinal Cord.
- Spinal Nerves.
- Transverse section of the spinal cord at the mid cervical level showing the general arrangement of the ascending tracts on the right and the descending tracts on the left.
- Functional components of spinal nerve fibers .
- Components and branches of spinal nerves.
- Arterial supply and venous drainage of spinal cord.
- Histological Structure of Nerve.
- Dermatomes.
- Myotomes.
- Main Descending and ascending Main pathway .

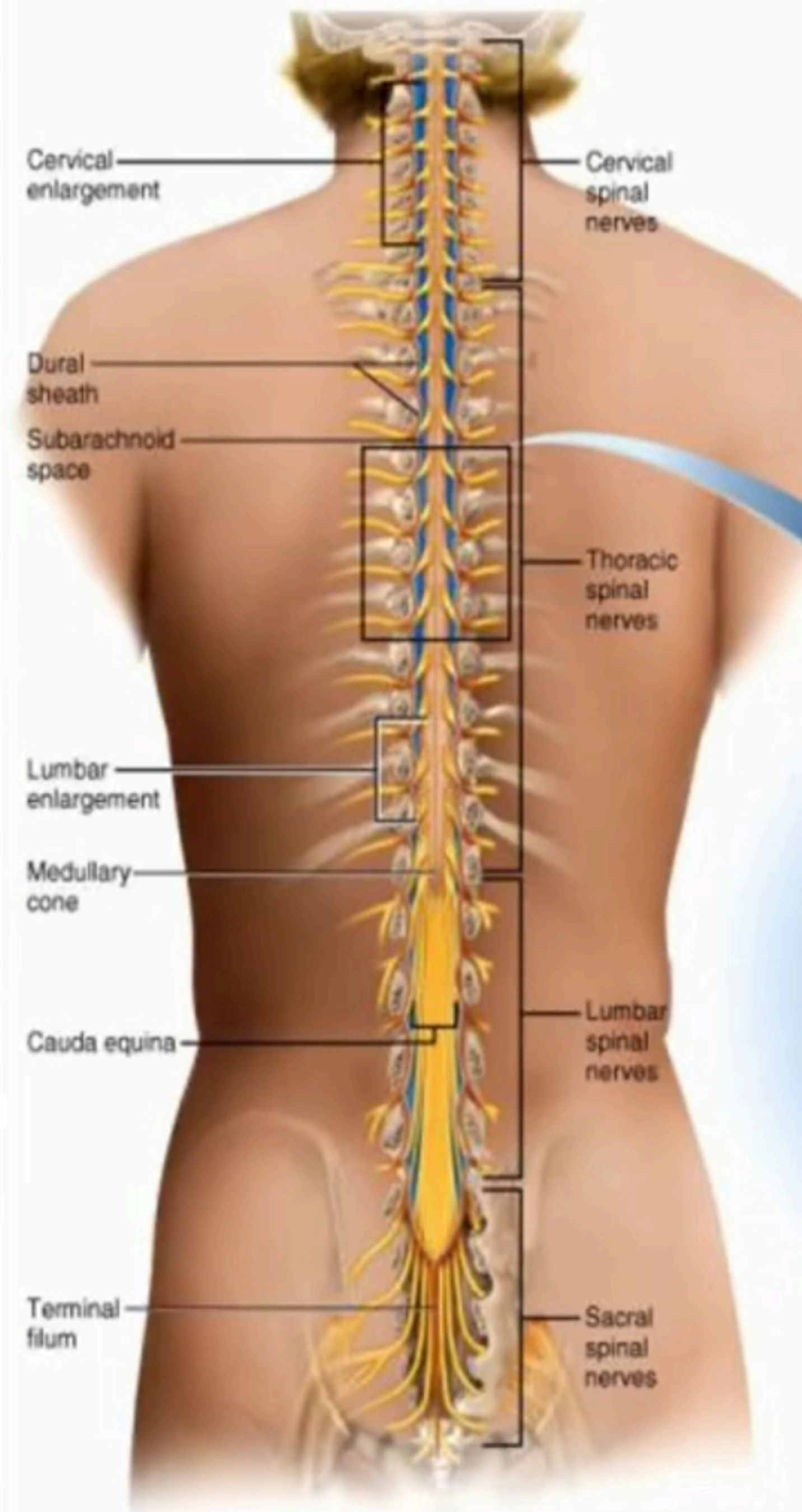
Spinal Cord

- It is **part of central nervous system** but located **extra cranially**.
- It is roughly **cylindrical** in shape.
- It begins **superiorly** at the **foramen magnum** in the **skull**.
- It is continuous **superiorly** with the **medulla oblongata** of the brain.

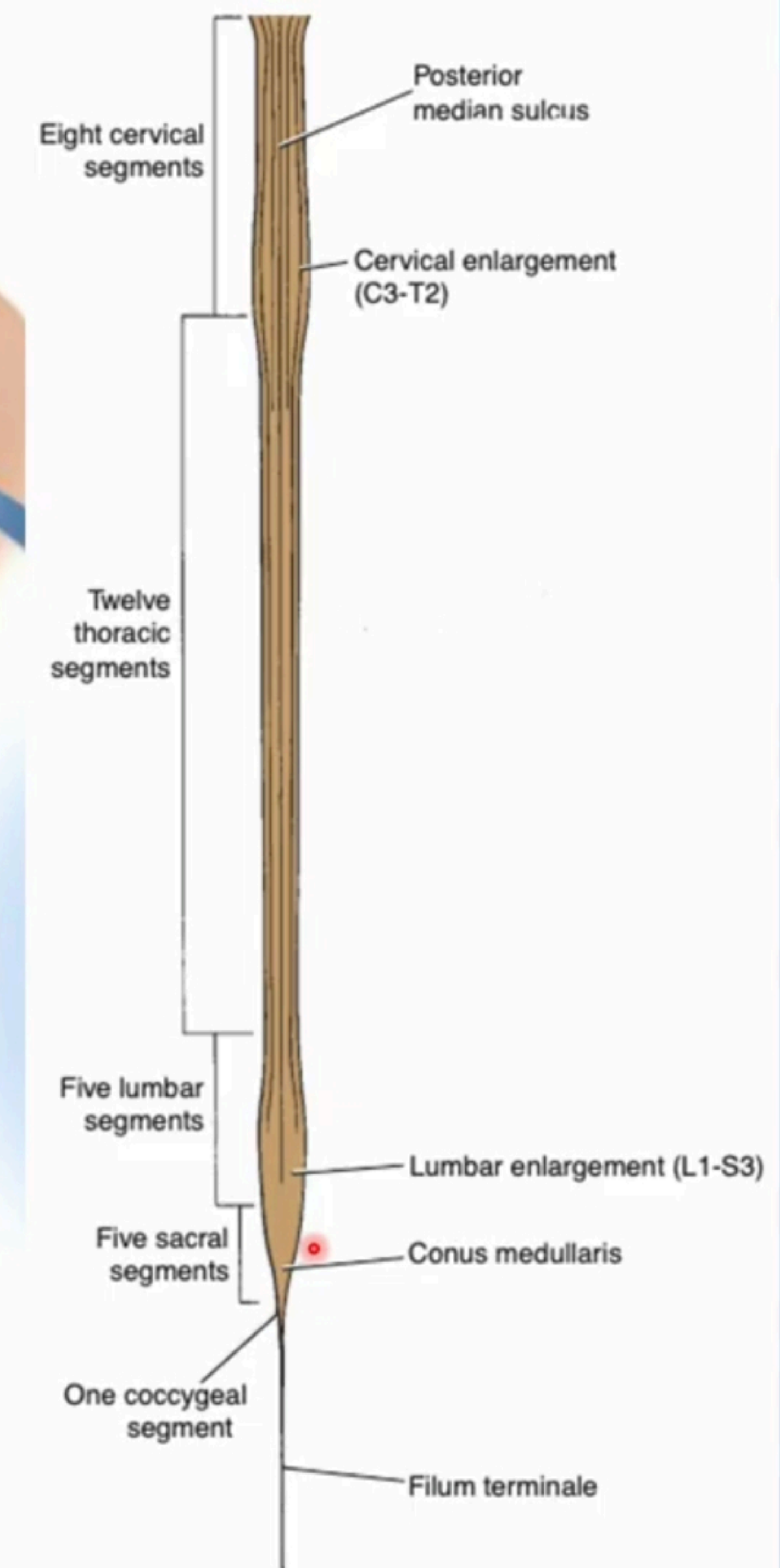


The Spinal Cord

- It terminates **inferiorly** called **conus medullaris**:
 - **In the adult** at the level of the lower border of the **first lumbar vertebra (L1)**.
 - **In the young child**, it is **relatively longer** that ends at the **upper border** of the **third lumbar vertebra**.
 - **In the fetal third month** ends **at coccyx**.
- It occupies the **upper two-thirds of the vertebral canal** of the vertebral column.

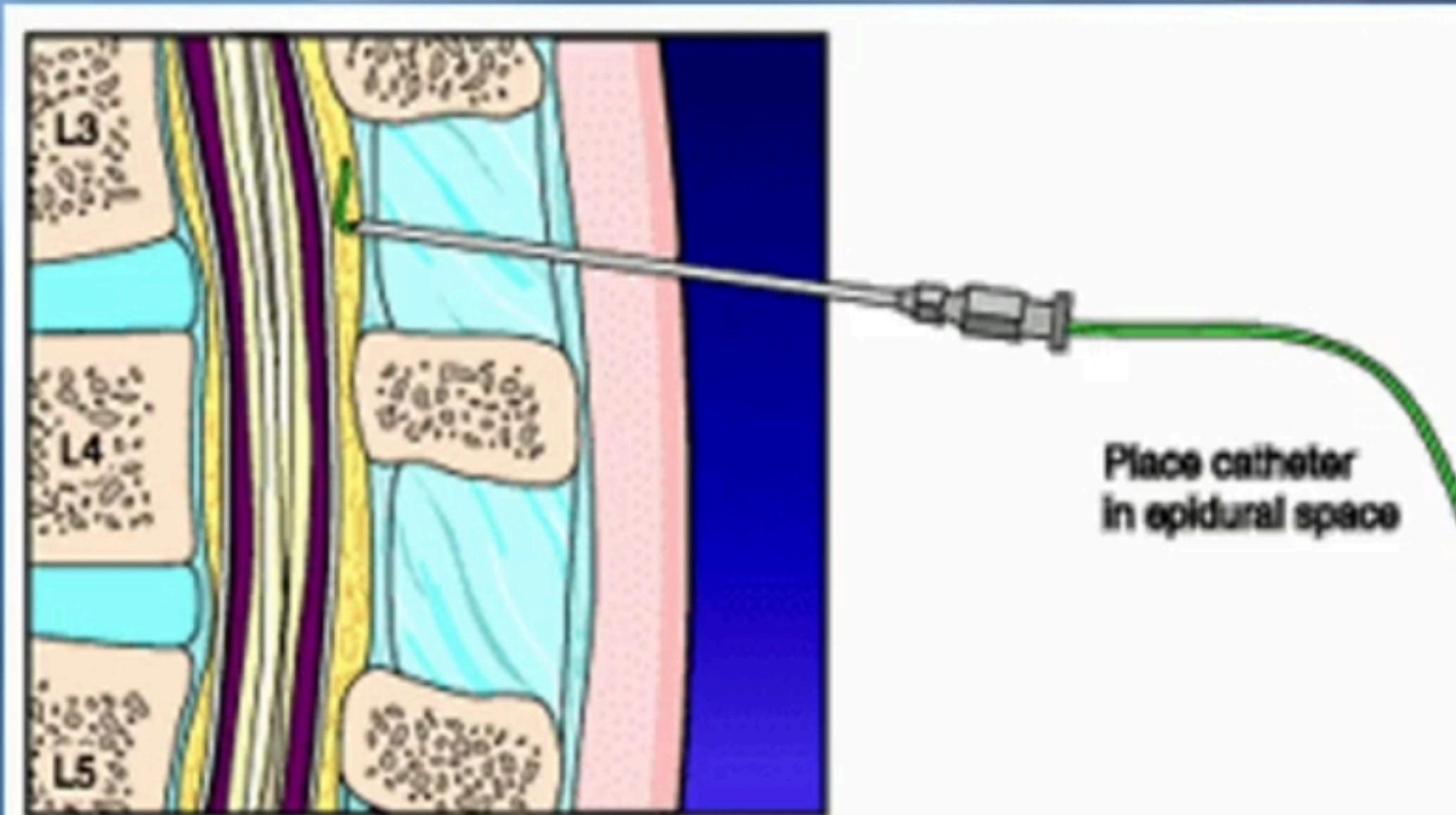
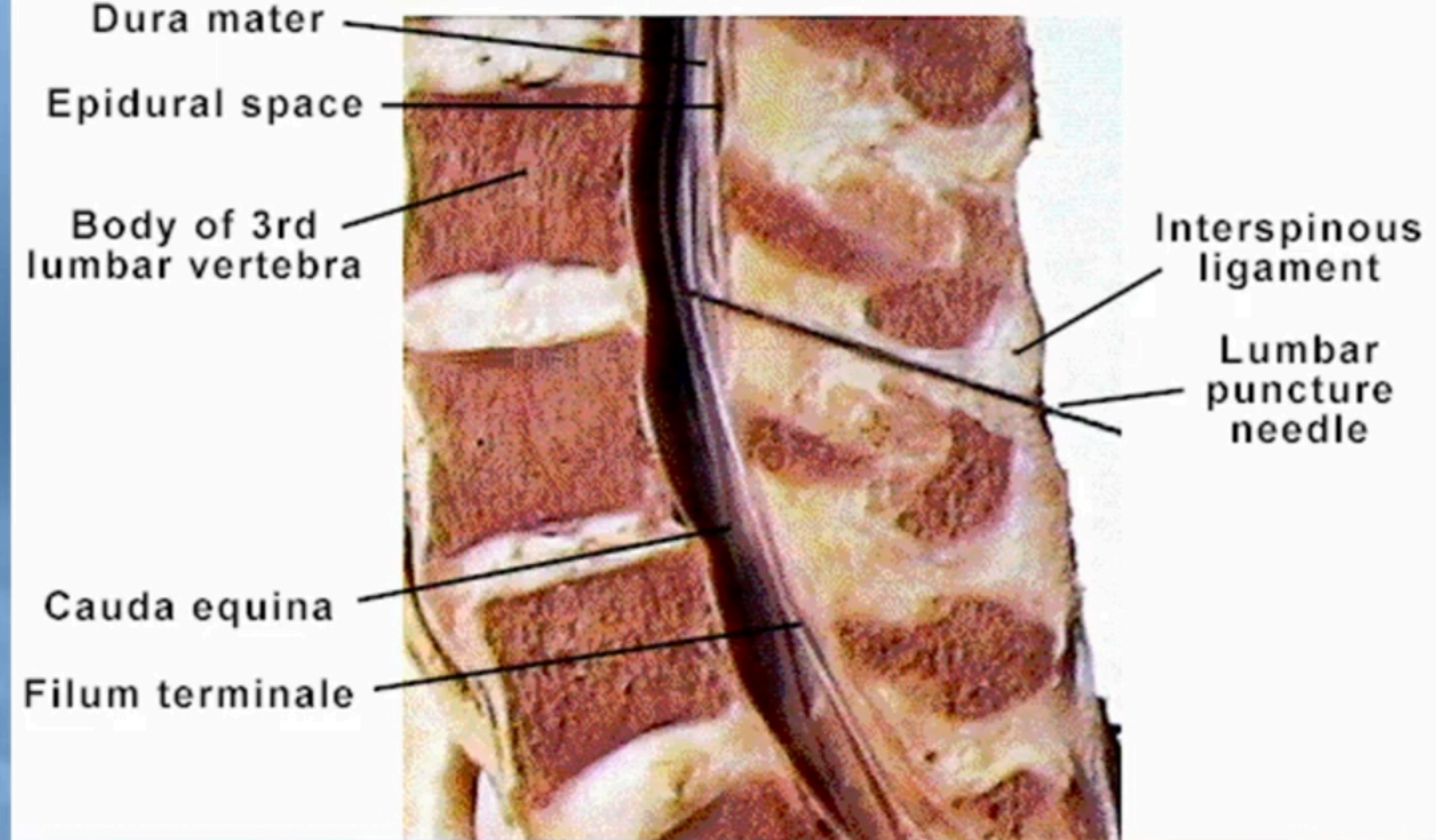


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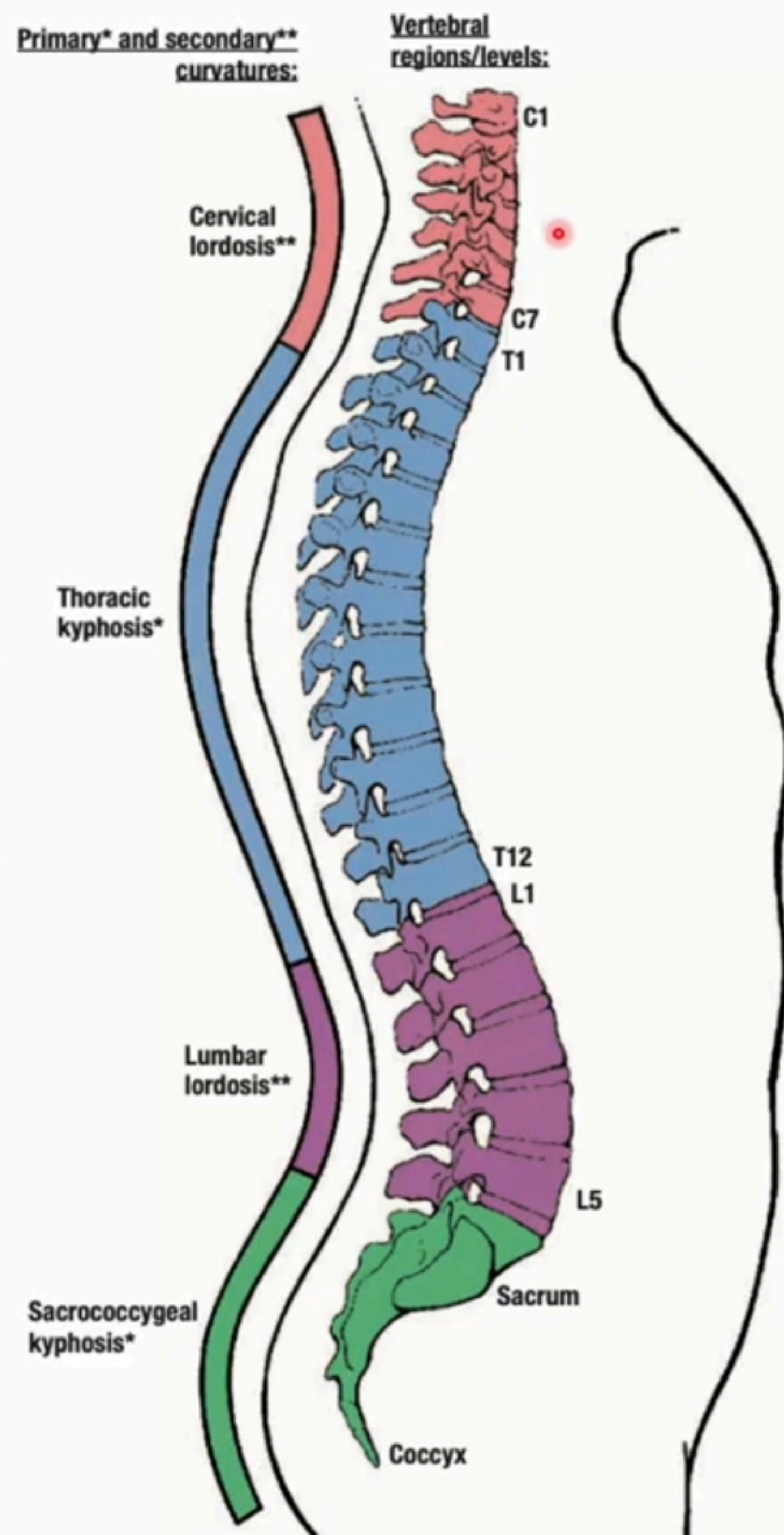
Lumbar Puncture (Spinal Tap)

- For clinical examination of CSF or administration of radiopaque dyes, drugs and sometimes anesthetics
- However: mostly “epidurals” for anesthetics



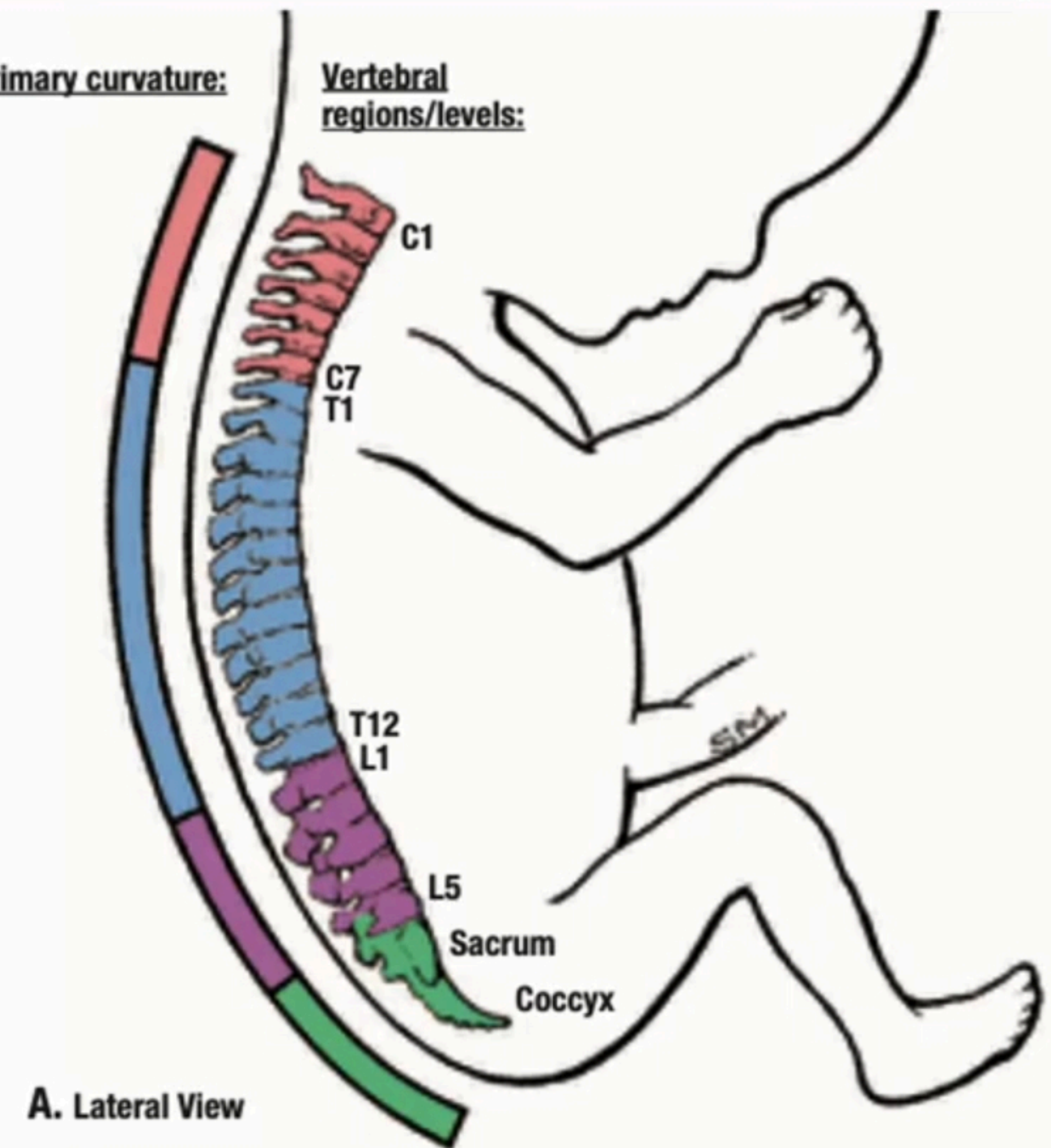
Curvatures of vertebral column deferate end of spinal cord between adult and infant and young child

➤ In the fetal third month ends at coccyx.



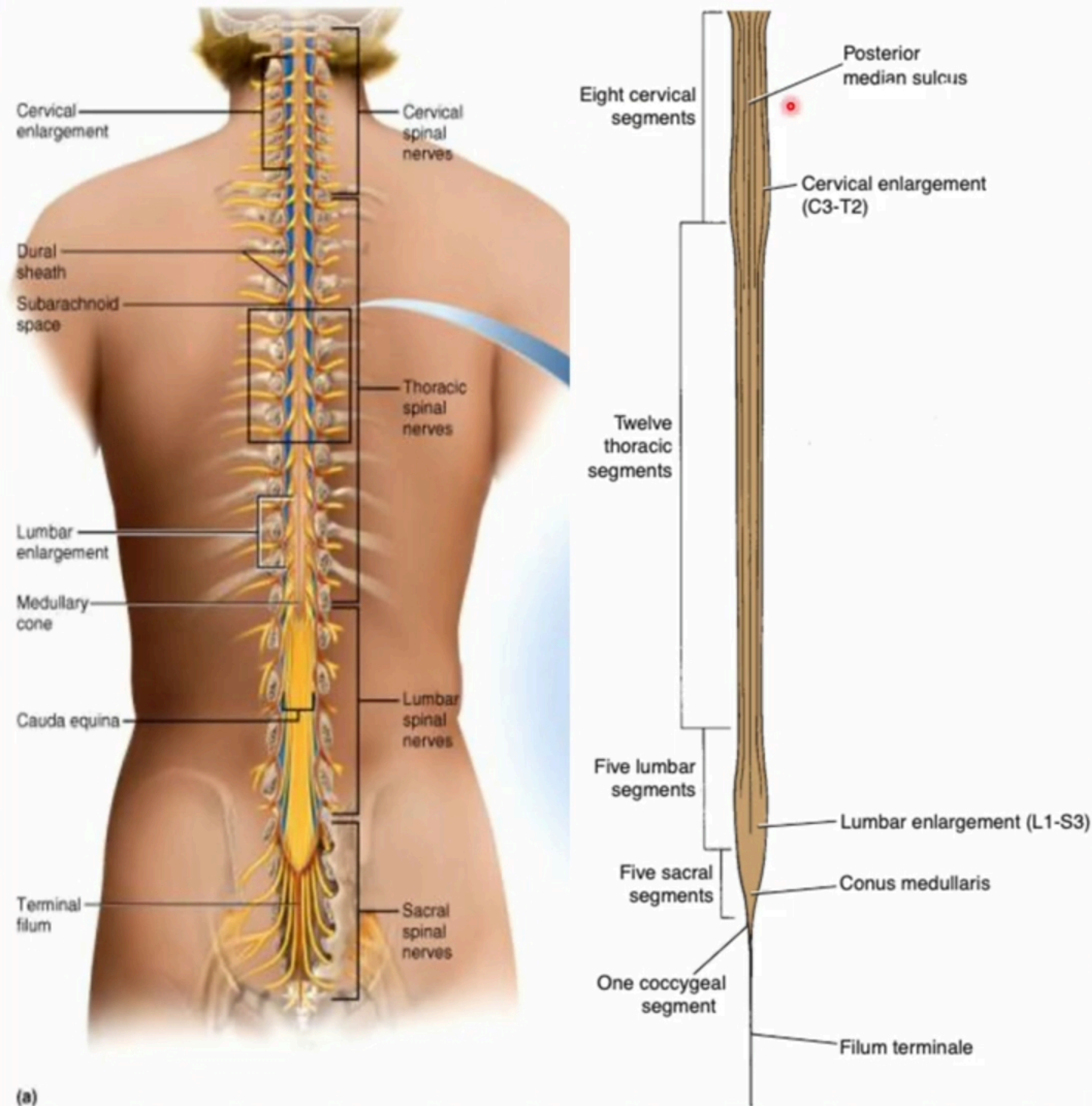
Primary curvature:

Vertebral regions/levels:



Spinal Cord

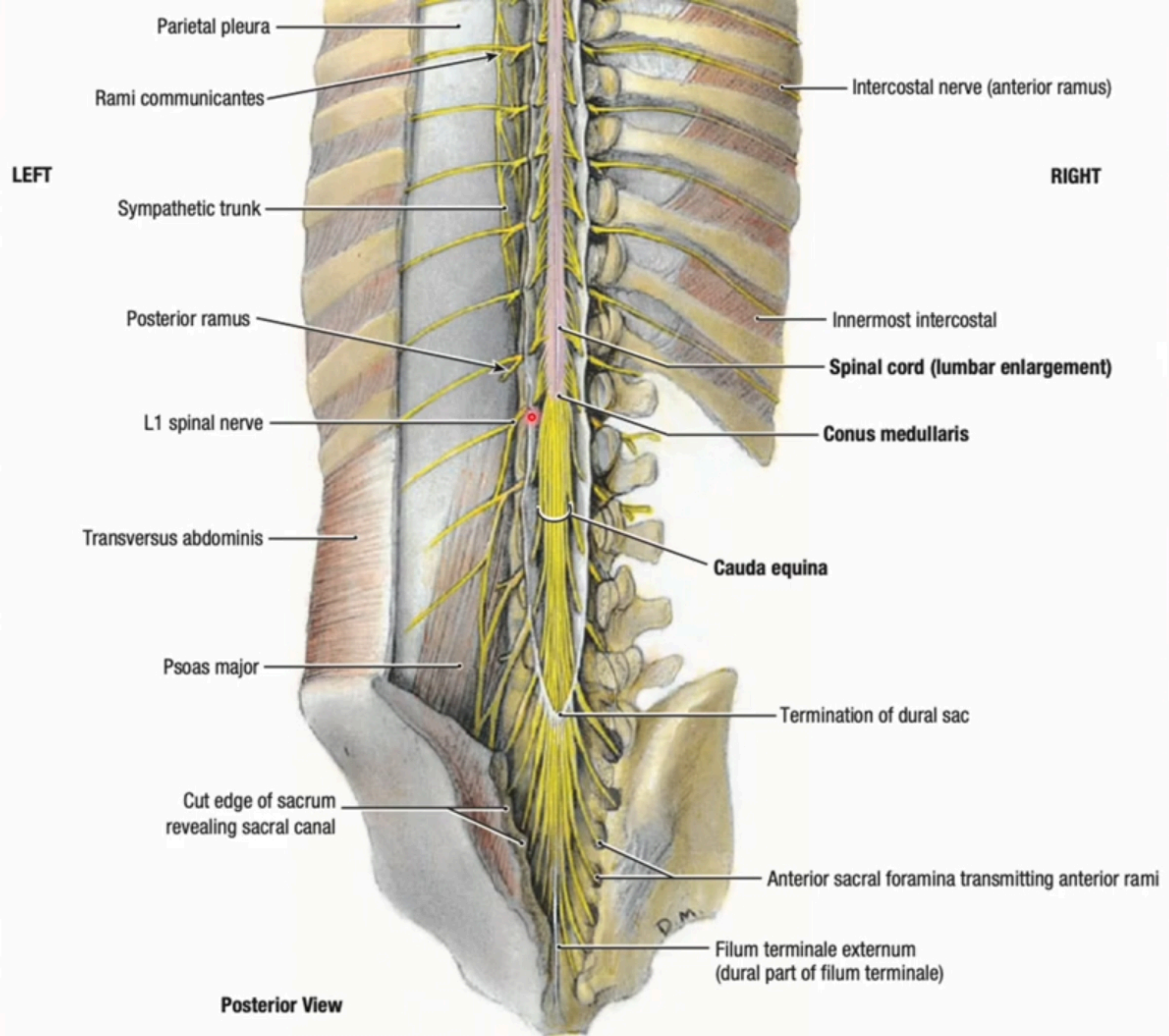
- The spinal cord is **fusiform enlarged**; the enlargements are referred to as **the cervical and lumbar enlargements**.
- **In the cervical region**, it gives origin to the **brachial plexus**.
- **In the lower thoracic and lumbar regions**, it gives origin to the **lumbosacral plexus**,



Spinal Cord

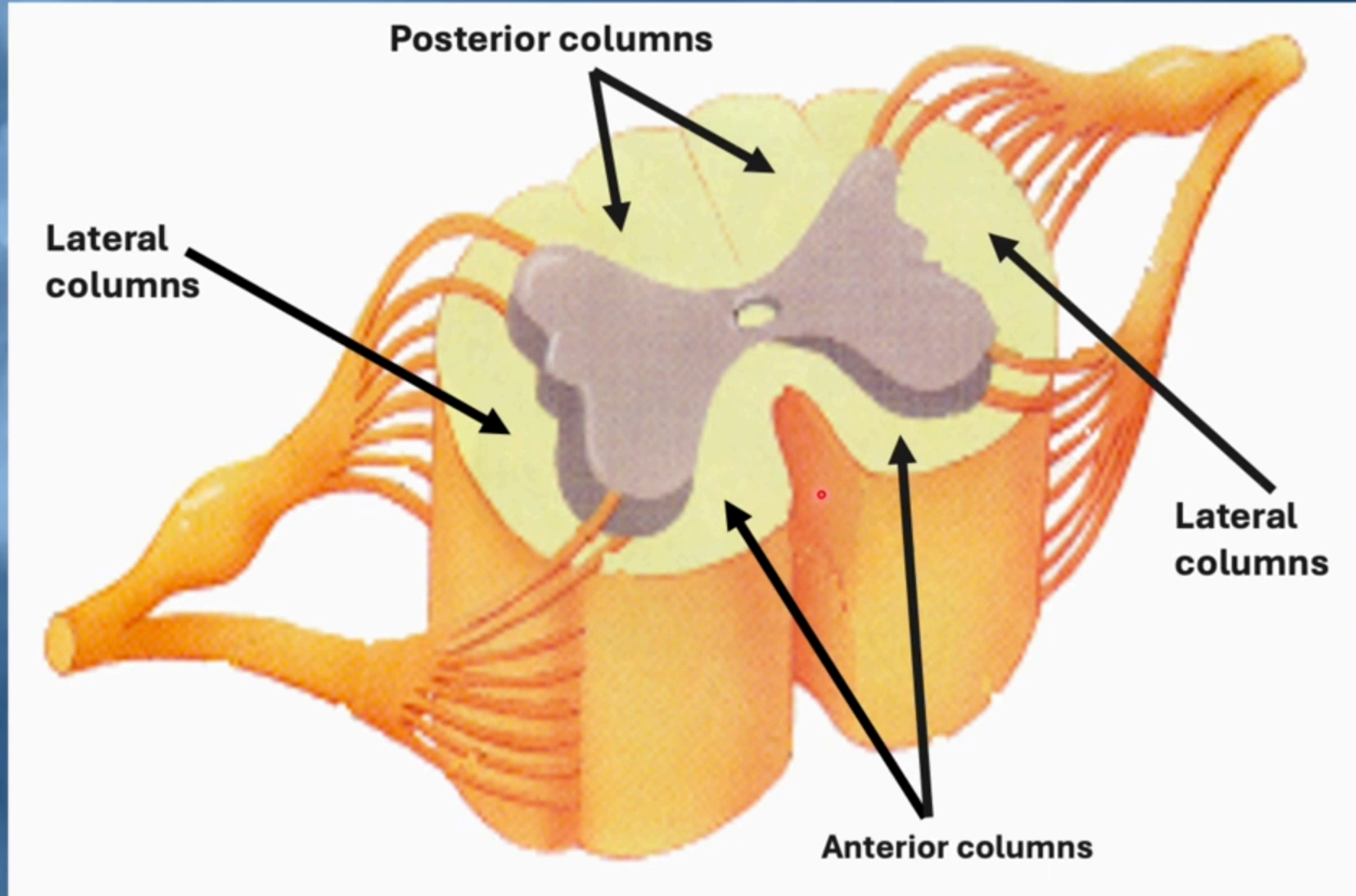
➤ **Inferiorly**, the spinal cord tapers off into **the conus medullaris**, from the apex of which a prolongation of **the pia mater, the filum terminal**, descends to be attached to the **posterior surface of the coccyx**.

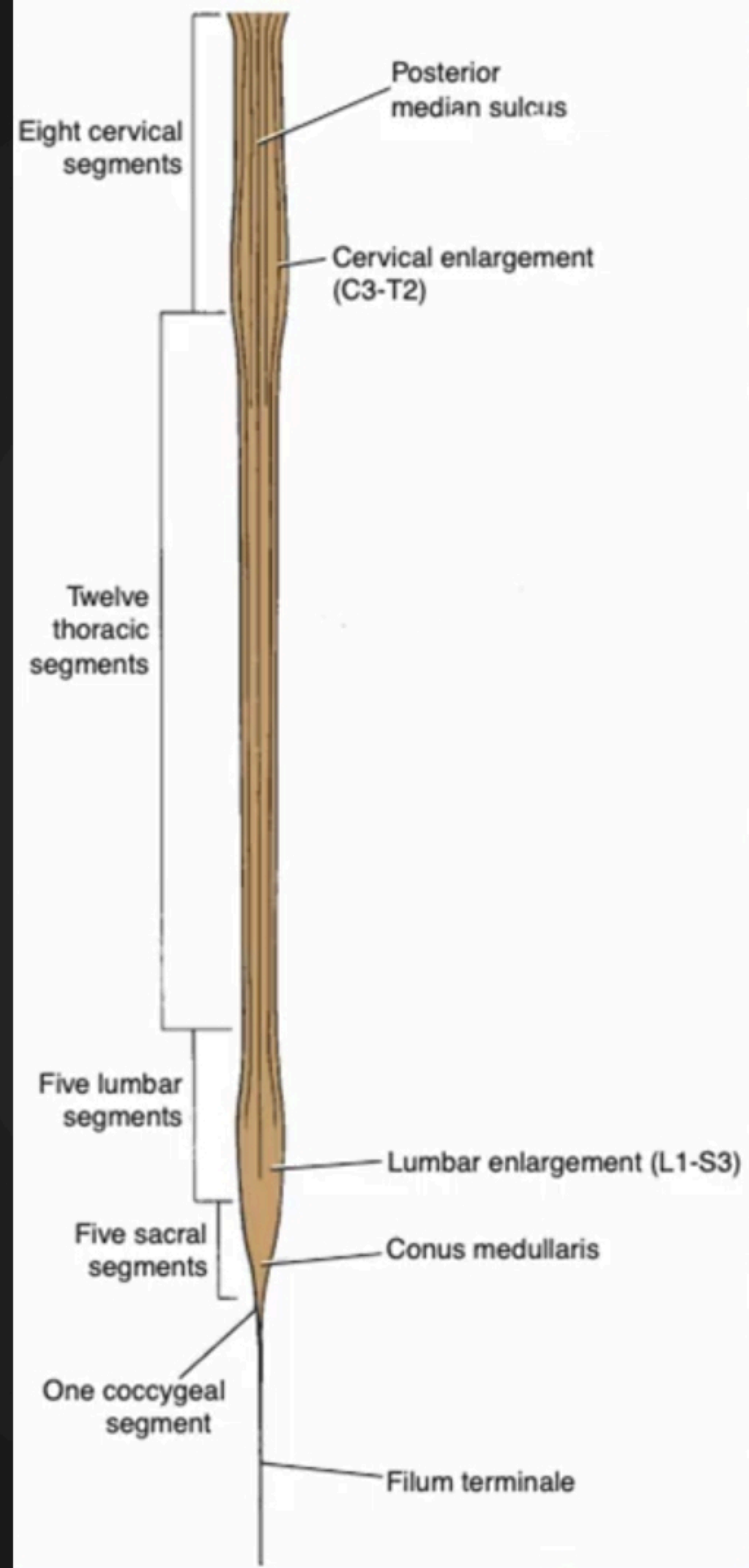
➤ The cord possesses a deep longitudinal fissure called **the anterior median fissure** in the midline anteriorly and a shallow furrow called the **posterior median sulcus** on the posterior surface.



Functions

- **Two-way conduction pathway** between the body and the brain.
- **Sensory and motor innervation** of entire body inferior to the head through the **spinal nerves**.
- **Major center for reflexes.**

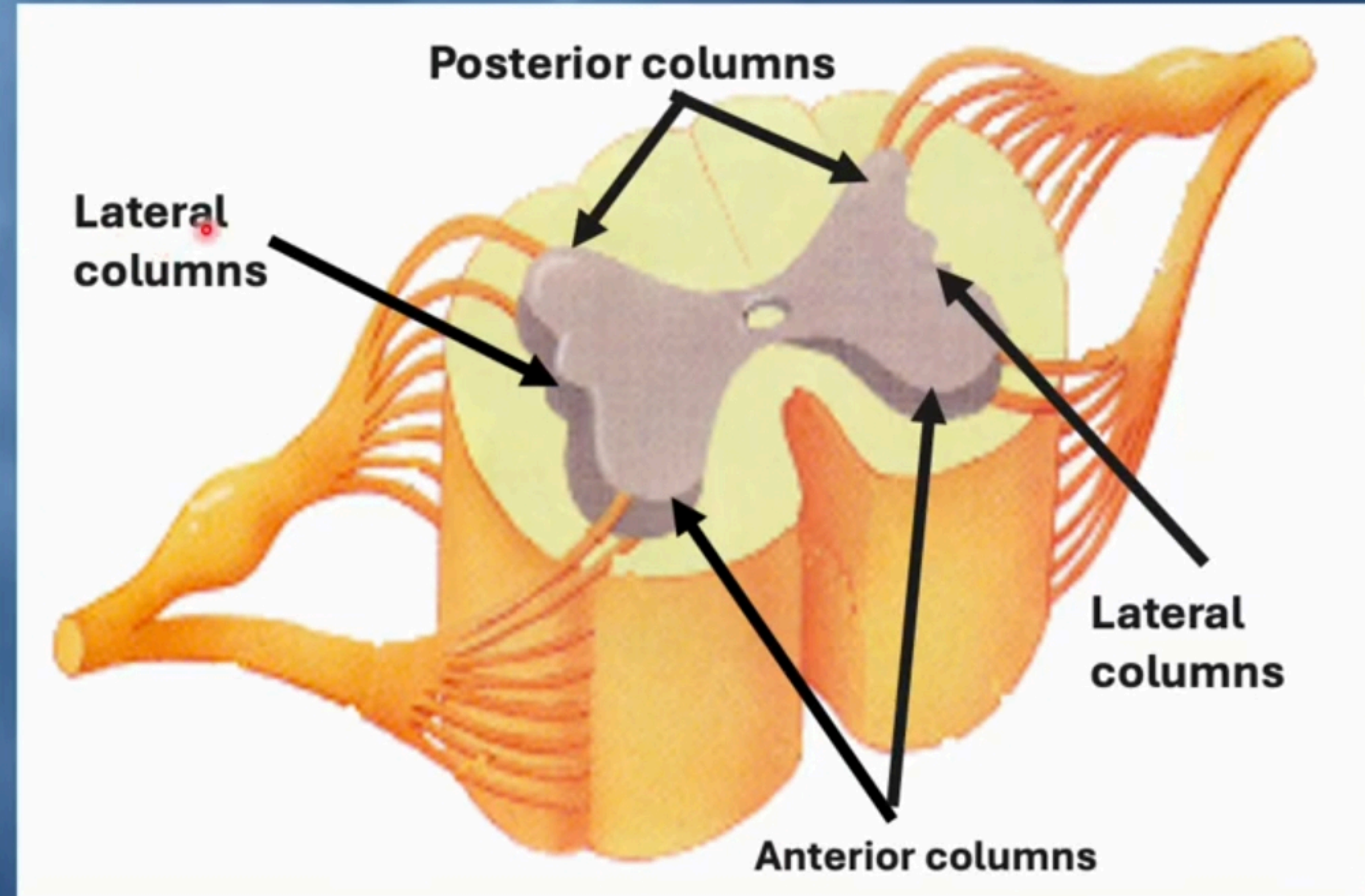




Spinal Cord

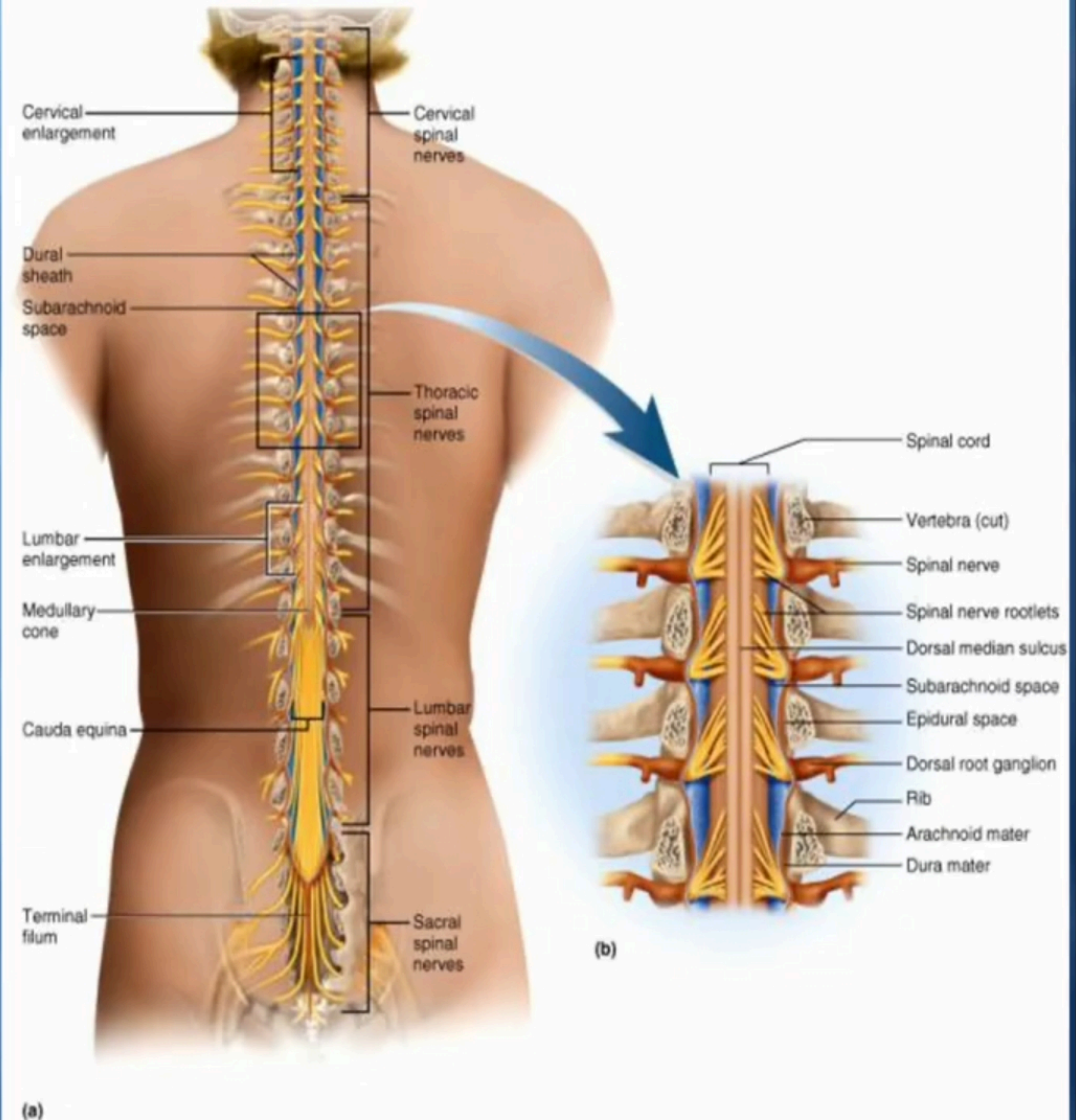
External Features

Internal Features



External Features

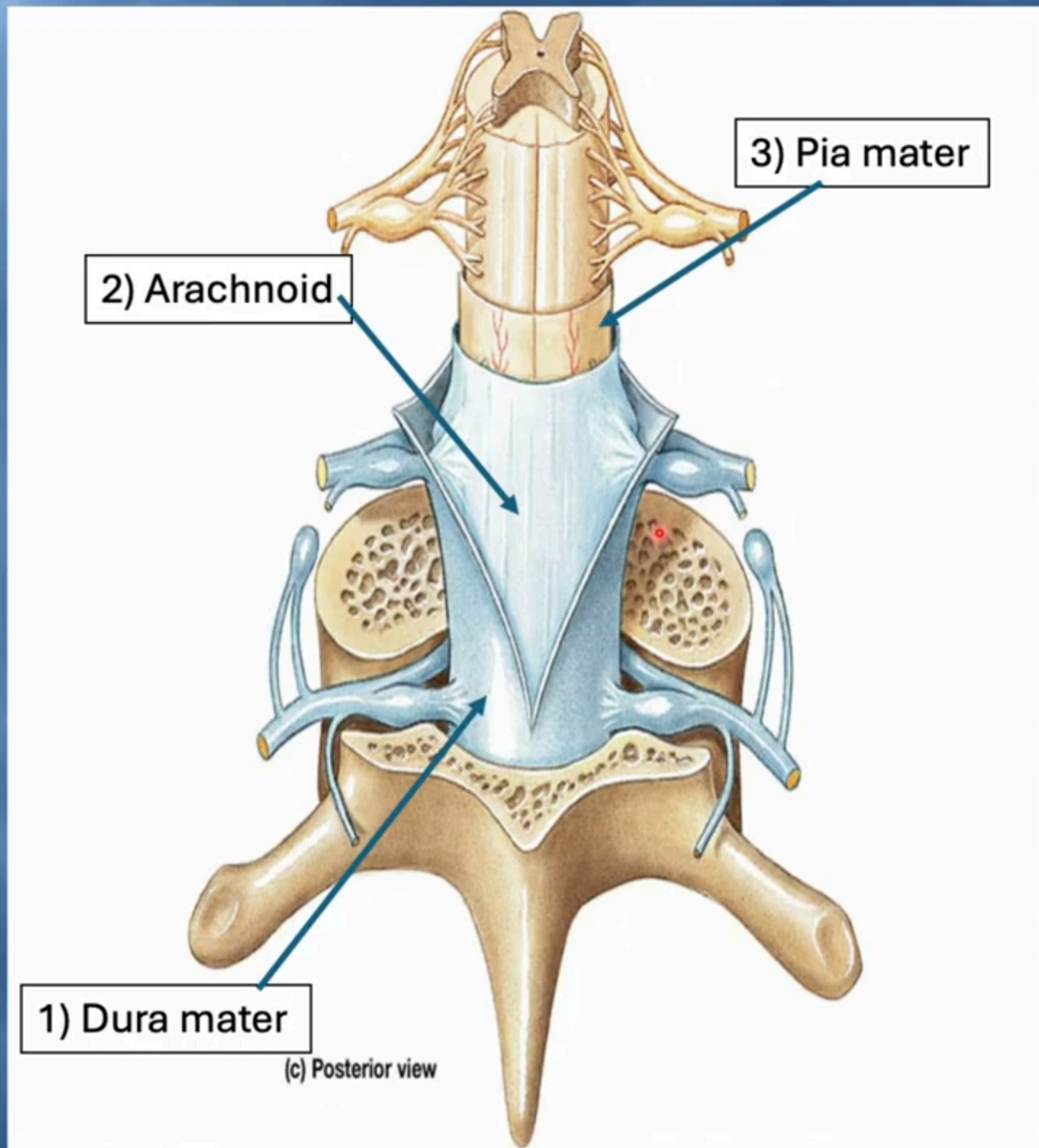
- **Cylindrical** in shape.
- **Length 45 cm**, Weight 35 grams, **Diameter** is about 1 cm
- **Surrounded by three meninges:**
 - Pia.
 - Arachnoid.
 - Dura mater.
- Appears segmented by **emergence of spinal nerves**



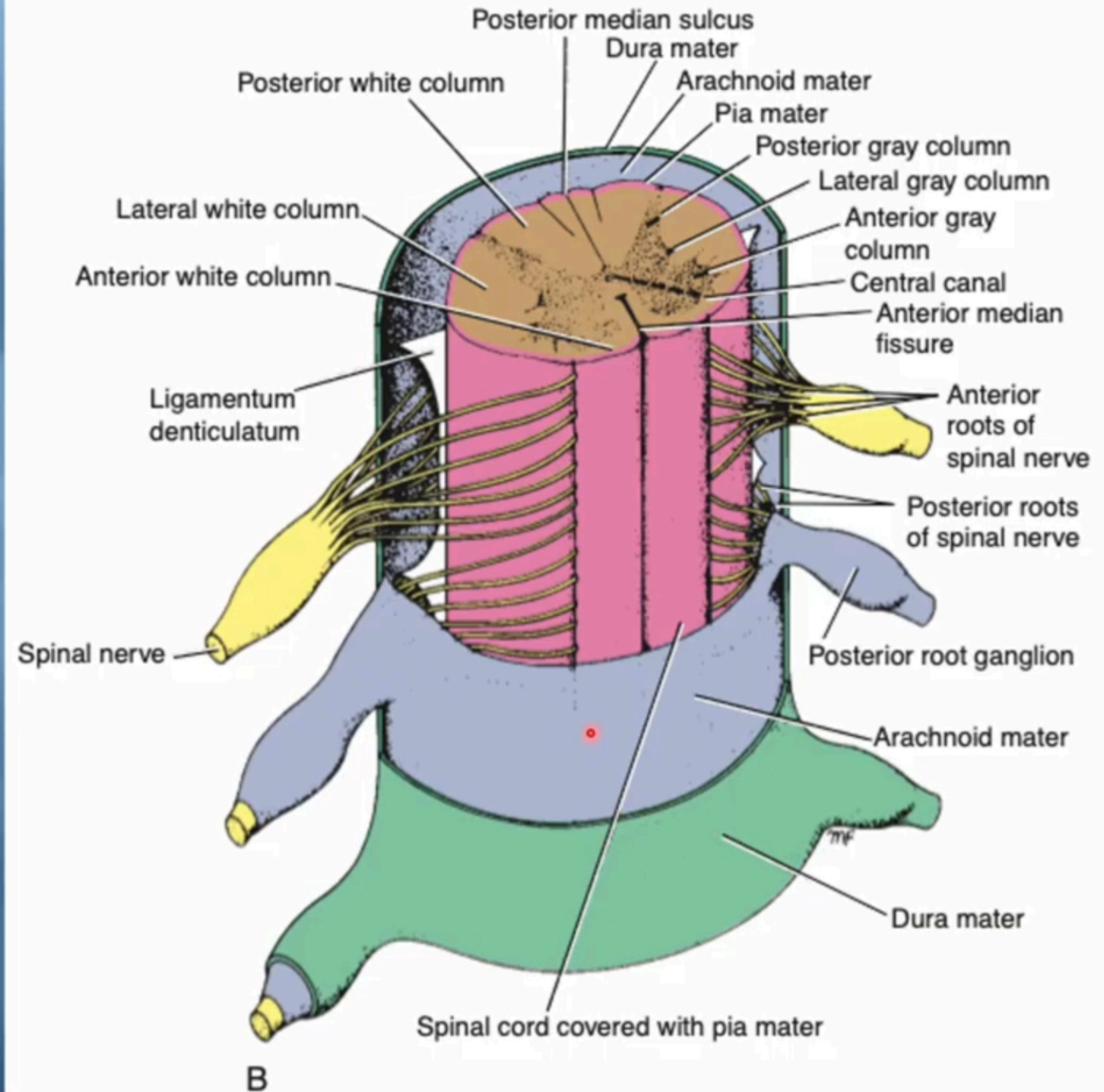
Spinal Meninges

Three membranes surround all of CNS:

- 1) **Dura mater** - "tough mother", strong.
 - 2) **Arachnoid meninx** - spidery looking, carries blood vessels, etc.
 - 3) **Pia mater** - "delicate mother", adheres tightly to surface of spinal cord
- **Filum terminale** - filamentous end of meninges, "tie-down"



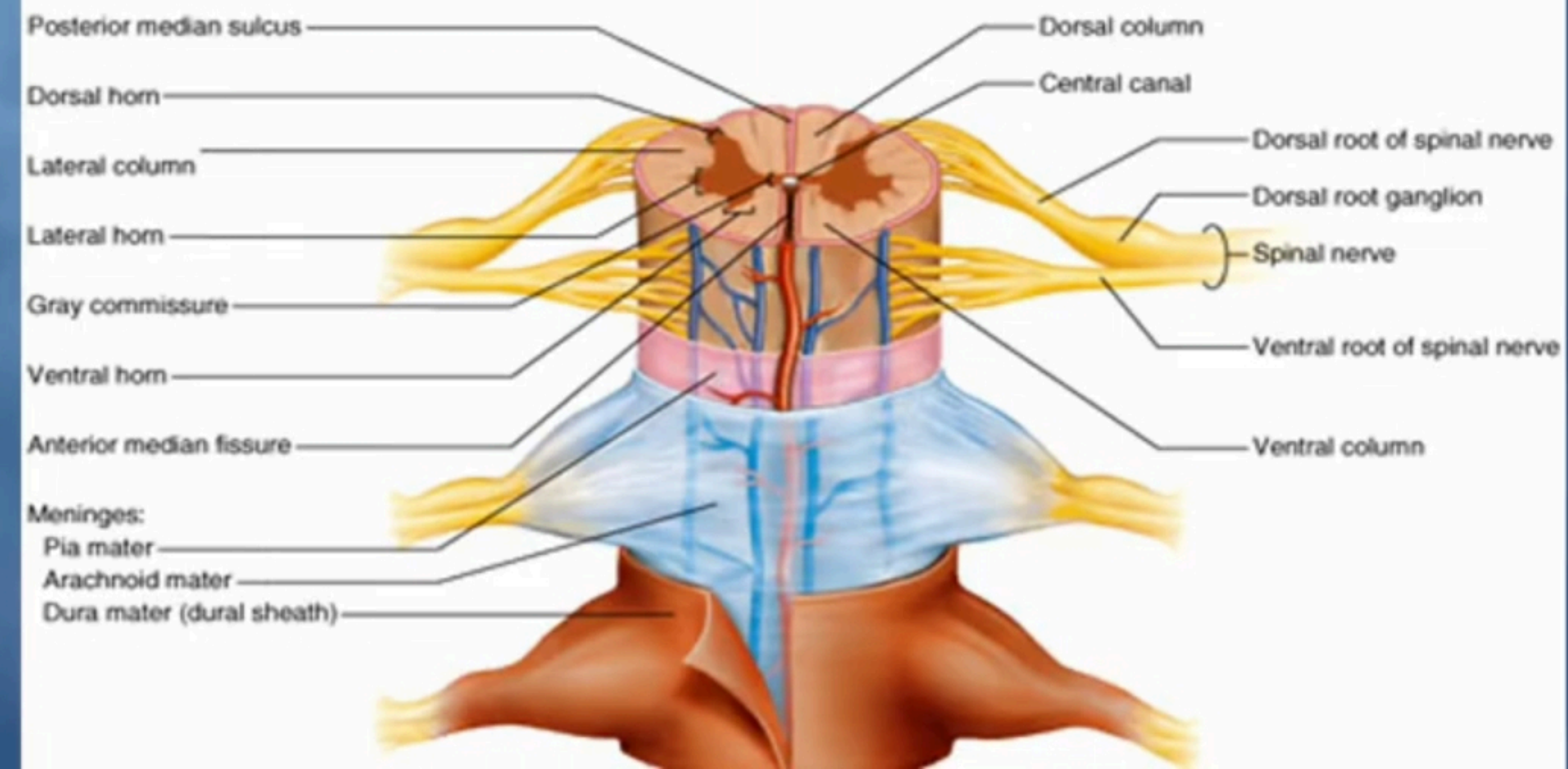
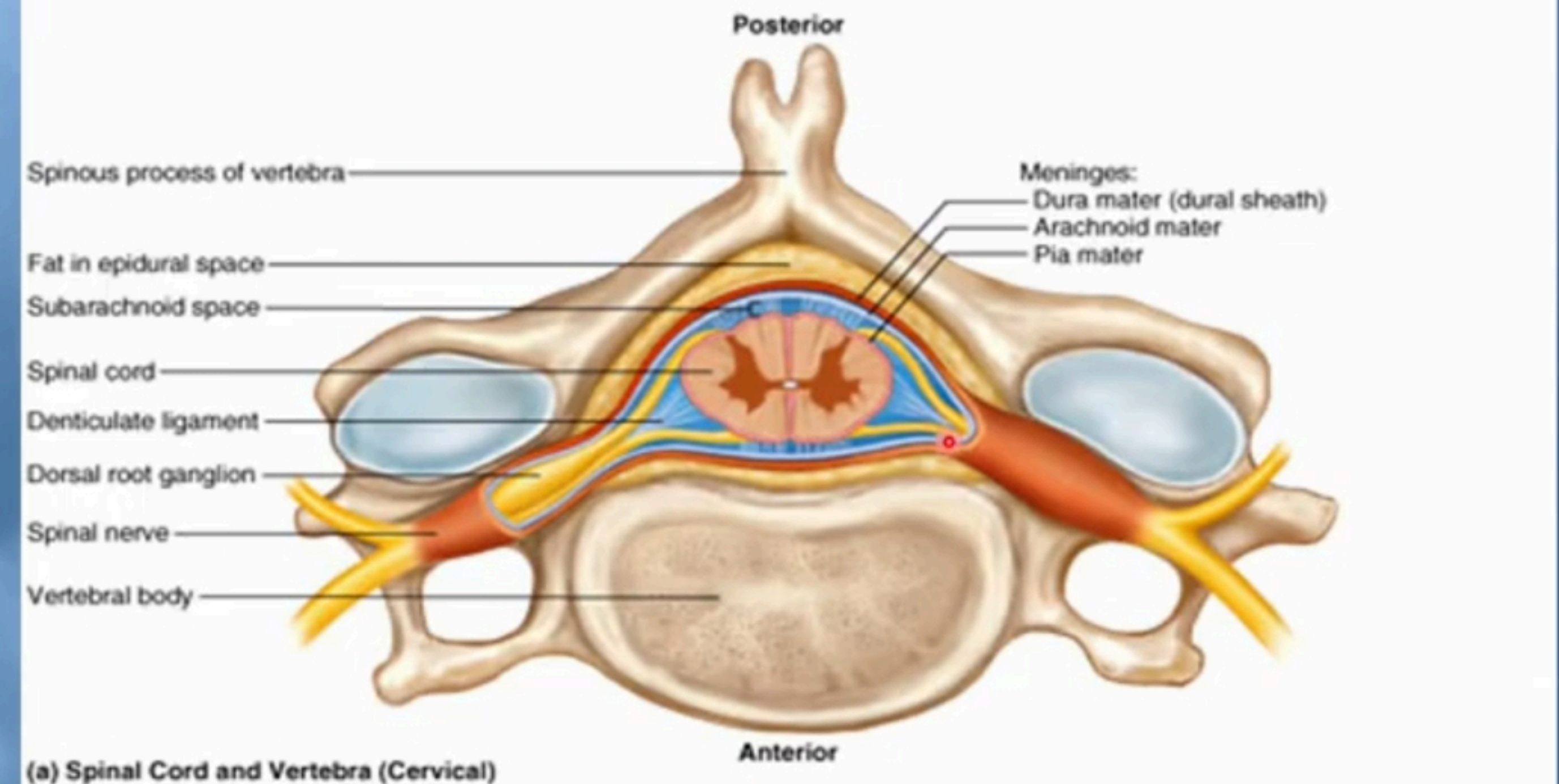
Spinal Meninges



Meninges And Spaces

1. Epidural space
2. Dura mater
3. Subdural space
4. Arachnoid mater
5. Subarachnoid space
6. Pia mater.

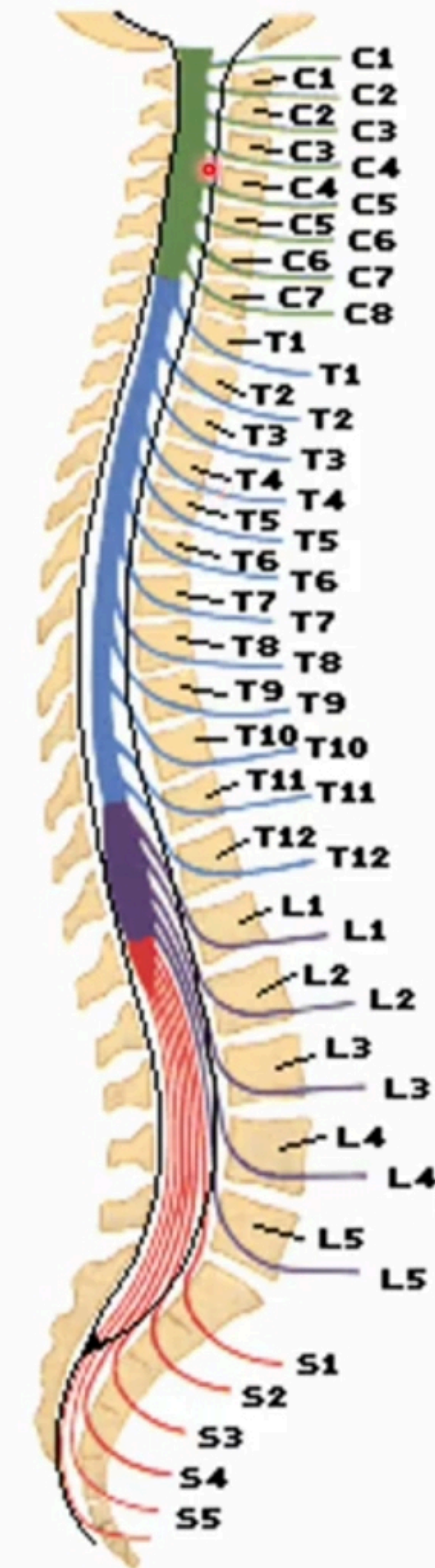
➤ Denticulate ligaments



Spinal Segments

Segments are:

- 8 Cervical.
- 12 Thoracic.
- 5 Lumbar.
- 5 Sacral.
- 1 Coccygeal.



Will be continue ...

